Real-World Design and Analysis of an AC Filter Circuit with Application and Resonance Insight

[a]Mohammed R. Awad

[b]Al-Aqsa University, Gaza, Palestine

[c]Advanced Electrical and Electronic Circuits (ENGS2304)

[d]Dr. Mohammed Naim

Department of Computer and Information Sciences

Faculty of Engineering

Second Semester 2026

Table of Contents

1. Theoretical Background

This section reviews the foundational concepts from Chapter 1 (Sinusoidal Steady-State Analysis) and Chapter 2 (AC Filter Circuits) that are needed to analyze the real-world filter presented later in the report.

1.1. Sinusoidal Steady-State

In electrical engineering, sinusoidal signals are of particular interest because most AC power sources, communication signals, and audio signals are sinusoidal or can be analyzed into sinusoids. A sinusoidal voltage is described by three parameters: amplitude Vp, angular frequency ω (or ordinary frequency f, with ω = 2πf), and phase φ, and is written $as\ v(t) = Vp \cdot cos(\omega t + \varphi)$ [1].

A linear system driven by a sinusoidal input reaches a steady state in which the output is also a sinusoid at the same frequency as the input, but generally with a different amplitude and a different phase shift [1]. This is the sinusoidal steady-state condition. Because frequency does not change, analysis can be carried out at a single frequency, which dramatically simplifies the math.

1.2. Phasors and Their Importance in AC Analysis

A phasor is a complex number that represents the amplitude and phase of a sinusoidal signal; it does not include frequency, since frequency is treated separately and assumed constant [1]. Phasors are introduced through Euler's identity:

$$e^{j\omega t} = cos(\omega t) + j \cdot sin(\omega t)$$

Using this identity, a cosine voltage $v(t) = Vp \cdot cos(\omega t + \varphi)$ can be written as $v(t) = \Re\{Vp \cdot e^{j(\omega t+\varphi)}\} = \Re\{Vp \cdot e^{j\omega t}\}$, where the phasor $V = Vp \cdot e^{j\varphi} = Vp\angle\varphi$ contains only the magnitude and phase, while the time variation $e^{j\omega t}$ is handled separately [1].

The benefits of phasor analysis are summarized in Chapter 1 [1]:

- Messy trigonometric functions are eliminated.
- Differentiation and integration are transformed into algebraic operations (multiplication and division by $j\omega$).

- The voltages and currents in any AC circuit can then be solved using Kirchhoff's laws and Ohm's law, which are used in DC circuits, but in the complex domain.

1.3. Impedance in the Phasor Domain (R, L, C)

Impedance Z is the voltage-to-current phase ratio of a component or network, V/I, and is measured in ohms (Ω). In general it is complex-valued [1]. So, what we learn in chapter 1 derives the impedance of the three passive elements as follows:

Resistor

In the time domain $v(t) = i(t) \cdot R$, so in the phasor domain $V = I \cdot R$, giving:

$$Z\_R = R \quad (real, \angle 0°)$$

Voltage and current remain in phase for a resistor [1].

Capacitor

From $i(t) = C \cdot dv/dt$ with $v(t) = Vp \cdot cos(\omega t + \varphi)$, differentiation gives $i(t) = \omega C \cdot Vp \cdot cos(\omega t + \varphi + 90°)$. In phasor form this becomes $I = j\omega C \cdot V$, so:

$$Z\_C = 1/(j\omega C) = (1/\omega C)\angle - 90°$$

The capacitor impedance decreases as frequency increases. Physically, current depends on how fast the voltage is changing — not on the voltage itself — so the current always leads the voltage by 90° in a capacitor [1]. Capacitors therefore pass high-frequency signals and block low-frequency ones.

Inductor

From $v(t) = L \cdot di/dt$ the analogous derivation gives $V = j\omega L \cdot I$, so:

$$Z\_L = j\omega L = (\omega L)\angle + 90°$$

Inductor impedance increases as frequency increases. The voltage depends on the rate of change of current, so the current lags the voltage by 90° [1]. Inductors allow low-frequency signals to pass through and block high-frequency signals.

These three relations are summarized by the mnemonic ELI the ICE man: in an inductor (L), voltage E leads current I; in a capacitor (C), current I leads voltage E [1].

1.4. What an AC Filter Does, and Its Types

A filter is an electrical circuit that selectively passes signals of specific frequency ranges and attenuates others [2]. Filters are crucial in radios (to isolate a desired channel and block others), in power supplies (to remove noise and ripple), in audio systems (to direct bass, midrange, and treble to the correct loudspeaker), and at the input of analog-to-digital converters (to prevent aliasing) [2].

Chapter 2 identifies four basic filter types [2]:

- Low-Pass Filter (LPF): Passes signals with frequencies lower than a cutoff frequency and attenuates frequencies higher than the cutoff.
- High-Pass Filter (HPF): Passes signals with frequencies higher than a cutoff frequency and attenuates frequencies lower than the cutoff.
- Band-Pass Filter (BPF): Passes signals within a certain frequency range and attenuates signals outside that range.
- Band-Stop Filter (BSF, also called notch filter): Attenuates signals within a certain frequency range and passes signals outside that range.

Filters are built from resistors (which set time constants in conjunction with capacitors), capacitors (which pass AC and block DC, with Z_C decreasing as frequency increases), and inductors (which store energy in a magnetic field, with Z_L increasing as frequency increases) [2].

1.5. Frequency Response and the Bode Plot

Linear systems are fully characterized by their frequency response — the way the system responds to sinusoids at different frequencies. The frequency response function is the ratio of the output phasor to the input phasor [2]:

$$H(j\omega) = Y(j\omega) / X(j\omega)$$

It is generally a complex-valued function of frequency containing both a magnitude |H(jω)| (the gain) and a phase ∠H(jω) (the phase shift). The standard graphical tool for displaying H(jω) is the Bode plot, which uses two stacked plots [2]:

- Magnitude plot — gain vs. frequency, in decibels: $20 \cdot log_{10}|H(j\omega)|$.

- Phase plot — phase shift vs. frequency, in degrees.

The frequency axis is logarithmic so that filter behavior is easy to read across many decades. A flat magnitude curve means the system passes that band unchanged, while a falling magnitude curve means the system attenuates (filters) those frequencies [2].

2. Real Filter Circuit Analysis: 2-Way Passive Loudspeaker Crossover (Audio System)

2.1. Identification of a Real Circuit

The 2-way passive loudspeaker crossover is one of the most widely manufactured AC filter networks in the world. It is mass-produced as a complete unit by all major amplifier component suppliers and is used in virtually all high-resolution home speakers, car audio systems, and professional amplifier units available on the market today. Concrete commercial examples that align with the analysis below include the Dayton Audio XO2W-2K/2.5K/3K series (second-order Linkwitz-Riley design, 4 or 8 ohms impedance, and ≈ 150W) and the Eminence PXB2 series (such as the PXB2:1K6 and PXB2:3K5, second-order Butterworth design, 8 ohms impedance, and up to 400W).

These crossover boards split the full-range signal coming from the amplifier into two bands and route each band to the appropriate driver: a low-pass section feeds the woofer, and a high-pass section feeds the tweeter.

2.2 Schematic and Component Values

2-way car stereo speaker crossovers

2-way home stereo speaker crossovers

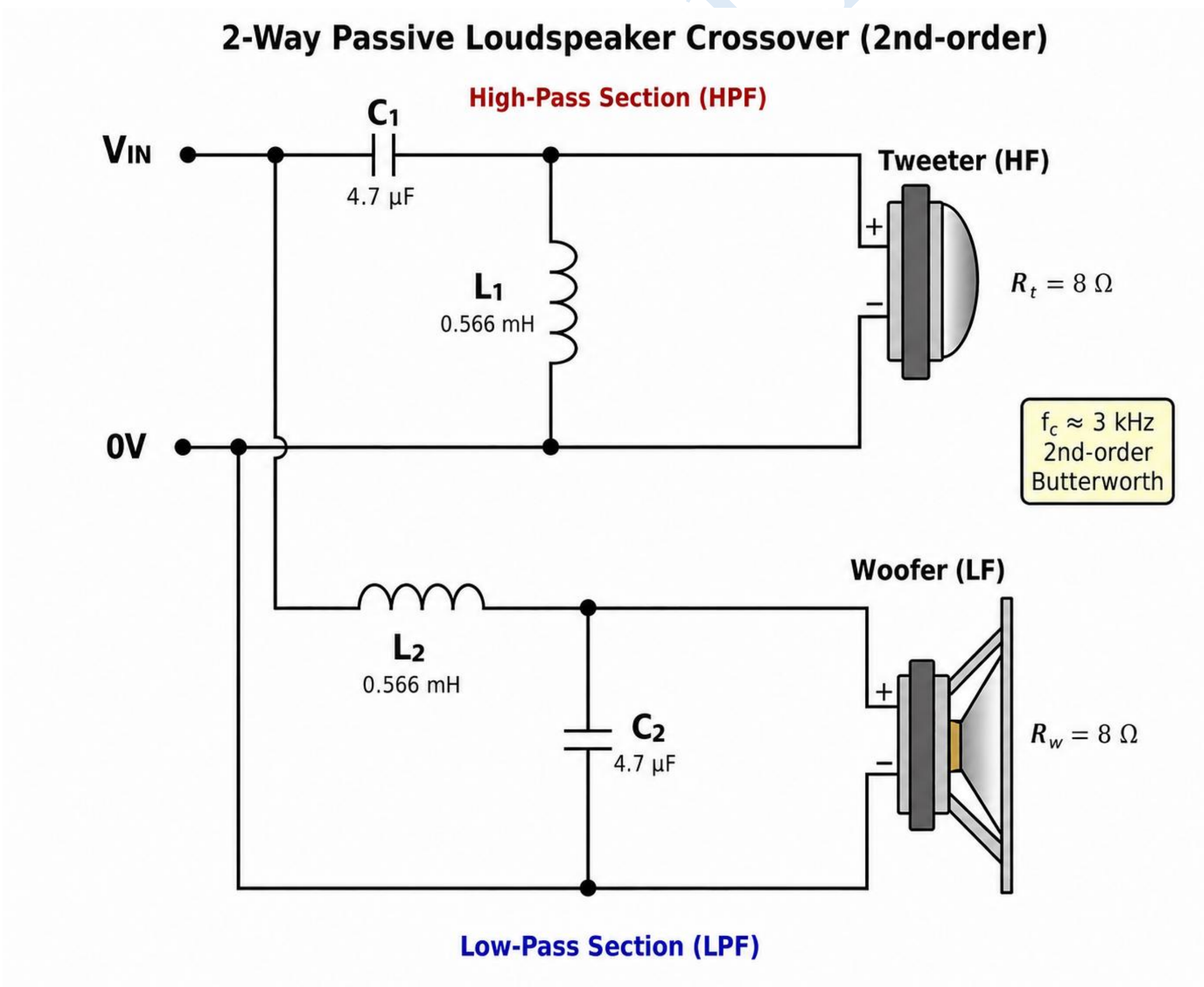


Figure 1 — Schematic of the 2-way passive crossover (2nd-order Butterworth, fc ≈ 3 kHz, 8 Ω drivers).

The four reactive components have the following roles:

- L_2 — series inductor in the low-pass arm. At low frequencies its reactance $X\_L = \omega L$ is small, so bass sound passes to the woofer; at high frequencies X_L is large, which blocks the high sound (treble).
- C_2 — shunt capacitor across the woofer. At high frequencies its reactance $X\_C = 1/(\omega C)$ is small, so it short-circuits residual treble to ground.
- C_1 — series capacitor in the high-pass arm. It blocks DC and low frequencies and passes treble through to the tweeter.
- L_1 — shunt inductor across the tweeter. Its low impedance at low frequencies short-circuits any bass to ground, protecting the tweeter.

Component values for the design example (chosen to match the commercial Dayton XO2W-3K): $L_1 = L_2 = 0.566\, mH, C_1 = C_2 = 4.7\, \mu F, with\, R\_w = R\_t = 8\, \Omega$ (nominal woofer and tweeter impedance).

2.3 Filter Type

The circuit in Figure 1 implements two filters operating in parallel on the same input:

- Low-pass filter (woofer arm) — series-L, shunt-C, 2nd-order, slope −40 dB/decade.
- High-pass filter (tweeter arm) — series-C, shunt-L, 2nd-order, slope +40 dB/decade.

Both arms share the same cutoff frequency *fc* and form a complementary pair: the power removed by the low-frequency filter from the woofer is the same power supplied by the high-frequency filter to the tweeter. This is the basic definition of a distribution network.

2.4 Transfer Functions H(jω)

Treating each driver as a resistive load (a simplification valid near the crossover frequency, away from the driver's natural resonance), the transfer function of the low-pass arm is derived using a voltage divider with $Z\_L = j\omega L$ in series and the parallel combination Z_C ∥ R as the output:

$$H\_LP(j\omega) = V\_woofer \,/\, V\_in = 1 \,/\, (1 + j\omega(L/R) + (j\omega)^2 \cdot LC)$$

By symmetry, the high-pass arm gives:

$$H\_HP(j\omega) = V\_tweeter / V\_in = (j\omega)^2 \cdot LC / (1 + j\omega(L/R) + (j\omega)^2 \cdot LC)$$

Comparing the LPF transfer function with the canonical second-order form $1 / (1 + s/(\omega_0 Q) + s^2/\omega_0{}^2)$ yields the two characteristic parameters of the filter:

$$\omega_0 = 1/\sqrt{(LC)} \quad and \quad Q = R \cdot \sqrt{(C/L)}$$

2.5 Cutoff Frequency

The cutoff frequency (angle) at -3 dB for a second-order LC filter is equal to its natural frequency $\omega_0/(2\pi)$:

$$fc = 1 / (2\pi \cdot \sqrt{(LC)})$$

Substituting the design values L = 0.566 mH and C = 4.7 μF:

$$fc = 1 / (2\pi \cdot \sqrt{(0.566 \times 10^{-3} \times 4.7 \times 10^{-6})}) = 1 / (2\pi \cdot \sqrt{(2.66 \times 10^{-9})}) \approx 3{,}086\, Hz$$

Therefore, the selected components produce a crossover frequency of approximately 3.09 kHz, which falls precisely within the typical frequency transition range between 2-way woofer-to-tweeter handover band of 2 and 3.5 kHz used in commercial two-way amplifiers.

Quality factor check:

$$Q = 8 \cdot \sqrt{(4.7 \times 10^{-6} / 0.566 \times 10^{-3})} = 8 \times 0.0912 \approx 0.73$$

This is essentially the Butterworth value $Q = 1/\sqrt{2} \approx 0.707$, confirming a maximally-flat passband response — the standard alignment used by Eminence in its PXB2 series.

2.6 Bode Plot

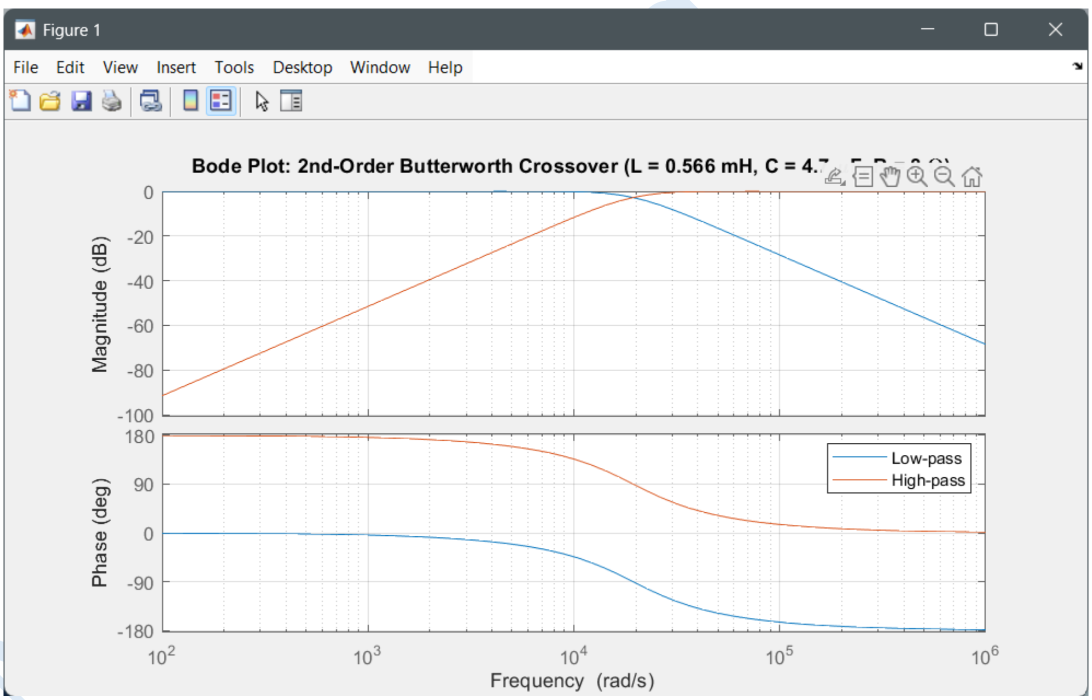
Editor - C:\Users\AL\Documents\MATLAB\HW2.m

HW1.m HW2.m

```matlab
% 2nd-Order Butterworth Crossover Bode Plot
% L = 0.566 mH, C = 4.7 uF, R = 8 ohms

clear; clc; close all;

L = 0.566e-3;      % Henry
C = 4.7e-6;        % Farad
R = 8;             % Ohm

s = tf('s');

% Low-pass LC crossover transfer function
H_lp = 1 / (L*C*s^2 + (L/R)*s + 1);

% High-pass LC crossover transfer function
H_hp = (L*C*s^2) / (L*C*s^2 + (L/R)*s + 1);

% Bode plot
figure;
bode(H_lp, H_hp);
grid on;

title('Bode Plot: 2nd-Order Butterworth Crossover (L = 0.566 mH, C = 4.7 \muF, R = 8 \Omega)');
legend('Low-pass','High-pass');

% Crossover frequency
fc = 1 / (2*pi*sqrt(L*C));
fprintf('Crossover frequency = %.2f Hz\n', fc);
```

Command Window

```
Crossover frequency = 3085.77 Hz
>>
```



In a cleaner view:

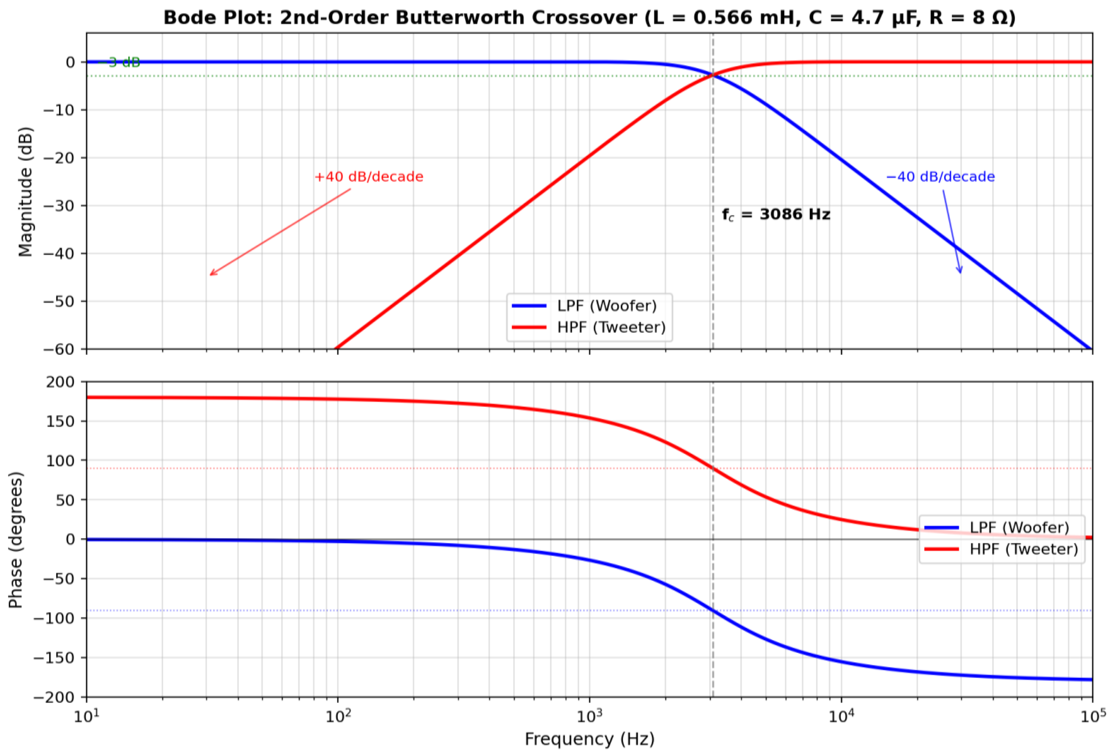


Figure 2 — A cleaner view of Bode plot of the 2-way crossover. Top: magnitude. Bottom: phase. Both filters cross at fc ≈ 3 kHz, where each is approximately −3 dB.

Key observations from Figure 2:

- In the LPF passband (f ≪ *fc*), |H_LP| ≈ 0 dB; in the LPF stopband (f ≫ *fc*), the value drops by −40 dB/decade (−12 dB/octave), which is the characteristic drop rate of any second-order filter[3].
- The HPF is the inverse: it rises by +40 dB/decade below *fc* and remains constant at 0 dB above *fc*.
- Both filters cross at exactly *fc* and at exactly −3 dB, with the HPF leading the LPF by 180° in phase. This 180° difference is why the tweeter must be connected with reversed polarity in a 2nd-order passive crossover so that the filter outputs combine constructively at *fc* rather than cancel each other out.

- The phase of the LPF passes through −90° at *fc* and approaches to −180° much higher than *fc*; the HPF passes through +90° at *fc* and approaches to 0° at high frequency.

2.7 Phasor Relationships Between Voltage and Current

In the AC steady state the phasor relationships in each branch are governed by the impedance rules from Chapter 1:

- In the series inductors L_1 and L_2: $V\_L = j\omega L \cdot I\_L$, so the inductor voltage leads the inductor current by 90° at every frequency [1].
- In the shunt capacitors C_1 and C_2: $I\_C = j\omega C \cdot V\_C$, so the capacitor current leads the capacitor voltage by 90°.
- At low frequency ($f \ll fc$): the LPF inductor presents almost zero impedance, so current flowing into the woofer is in phase with the source voltage — exactly as expected for a resistive load.
- At high frequency ($f \gg fc$): the HPF capacitor presents almost zero impedance, so the tweeter again behaves resistively.
- At $f = fc$: the inductive reactance ωL and the capacitive reactance 1/(ωC) are equal in magnitude and opposite in sign — a partial resonance condition that explains why the gain is −3 dB rather than 0 dB at the corner. This connects directly to the resonance discussion in Section 4.

3. Real-World Application

3.1 The System: A Multi-Driver Loudspeaker

A complete audio system, including the crossover, consists of an amplifier, the crossover network, and at least two audio transformers — a woofer (a large-diameter cone-shaped speaker, typically 5 to 12 inches) and a tweeter (a small dome or horn-shaped speaker, typically 0.5 to 1.5 inches). The signal path is simple: the amplifier sends a full-range audio signal (20 Hz to 20 kHz) to the speaker terminals, and the passive distribution circuit splits this signal into two frequency bands within the speaker enclosure. Each band is then directed to the appropriate speaker for reproduction[4].

This architecture is dominant across three application domains [4][5]:

- Home hi-fi: Two-way speakers, whether rack-mounted or floor-standing (such as KEF, ELAC, Polk, JBL, and Klipsch), typically operate at a crossover frequency of 1.8–3 kHz using second- or third-order passive frequency separation circuitry.
- Car audio system: Component kits (a 6.5-inch mid-range/low-frequency tweeter plus a high-frequency tweeter, sold with passive frequency separation circuitry) typically operate at a crossover frequency of 3–3.5 kHz with a 12 dB/octave gradient.
- Professional audio system: Professional speakers operate at a crossover frequency of 1.5–2.5 kHz between a 12/15-inch low-frequency tweeter and a horn-type compression tweeter, often using fourth-order Linkwitz-Riley active filters, which are implemented in digital signal processors.

3.2 What the Filter Passes and What It Blocks

With the design from Section 2 ($fc \approx 3$ kHz):

- The low-frequency pass-through arm to the low-frequency amplifier (woofer) passes the sound from 20 Hz up to about 3 kHz with minimal attenuation, then fades out at a rate of 12 dB/octave. At 6 kHz, the signal at the low-frequency amplifier is 12 dB lower than the input signal; at 12 kHz, it is 24 dB lower. Bass and low mid-frequencies (such as the sound of a bass drum, bass guitar, male vocals, and low piano notes) reach the low-frequency amplifier; however, cymbals, whistles, and airband content do not.
- High-pass arm to tweeter passes audio from around 3 kHz and above with minimal attenuation. Below 3 kHz, the frequency drops by 12 dB/octave, so at 1.5 kHz, the tweeter receives 12 dB less of the input signal, and at 750 Hz, it receives 24 dB less. Therefore, the tweeter reproduces cymbal sounds, brass harmonics, whistles, and string harmonics, while being shielded from the low and mid-range frequencies.[7]

3.3 Technical Justification — Why This Filtering Is Critical

3.3.1 Driver Protection

Tweeters are mechanically and thermally fragile. A typical dome tweeter has a maximum linear excursion (X_max) of only 0.2 – 0.5 mm and a voice coil so small (≈ 25 mm diameter, fine-gauge wire) that it cannot dissipate sustained low-frequency power without overheating [4][5]. Because driver excursion rises at 12 dB/octave below the driver's free-air resonance (typically 800 – 1500 Hz for tweeters), any unfiltered bass drives the dome past X_max almost instantly. A 50 W tweeter connected directly to a 200 W amplifier without a high-pass filter will fail mechanically (suspension torn, voice coil rubbing the magnet gap) within seconds of seeing real program material.

Woofers are mechanically robust but become inefficient and distorted above their useful range. Cone breakup, beaming (very narrow directivity), and intermodulation (Doppler) distortion become severe above ~2 kHz [4].

3.3.2 Driver Efficiency

Each driver is most efficient (highest sound pressure per watt of input) within its design band. Restricting each driver to about one decade of frequency lets the system deliver the maximum sound pressure for a given amplifier power. Empirically, professional audio engineers limit each driver to roughly three octaves and treat four octaves as the practical maximum before audible compromises appear [4].

3.3.3 Reduction of Intermodulation Distortion

When a single driver attempts to reproduce widely separated frequencies simultaneously, the cone motion from one frequency modulates the radiation of the other (Doppler / amplitude modulation), creating sidebands that are not in the source. Splitting the band between two drivers via the crossover essentially eliminates this effect.

3.4 Consequences If the Filter Did Not Exist

Symptom	Underlying mechanism
Tweeter blows almost instantly	Voice-coil overheating from low-frequency power; suspension over-excursion below the tweeter's free-air resonance.
Boomy, smeared midrange	Intermodulation (Doppler) distortion in the woofer when high frequencies are present at large cone excursion.
Lobing / comb-filtering	Two drivers radiating the same band from different acoustic centres interfere constructively in some directions and destructively in others.

Symptom	Underlying mechanism
Reduced effective SPL	Drivers operating outside their efficient band waste amplifier power as heat.
Shortened service life	Continuous thermal cycling and mechanical fatigue of voice coils and suspensions.

Table 1 — Failure modes that arise when no crossover filter is present.

In short, the crossover network is not a tonal refinement; it is the component that makes a multi-driver loudspeaker physically possible. Removing it transforms a hi-fi speaker into a self-destruction device within seconds of being connected to a serious amplifier.

4. Resonance Insight

Because the crossover circuit contains both inductors and capacitors, it is by definition a second-order circuit (the order of a circuit equals the number of independent energy-storage elements [6]) and is therefore capable of exhibiting resonance. This section connects the filter design of Section 2 with the resonance theory developed in Chapter 3.

4.1 Can the Circuit Exhibit Resonance?

Yes. From what we learned in Chapter 3, electrical resonance occurs whenever inductive reactance and capacitive reactance cancel:

$$X\_L + X\_C = 0 \quad \Rightarrow \quad X\_L = -X\_C$$

In the low-pass arm of the crossover, this condition is met when:

$$\omega_0 \cdot L = 1/(\omega_0 \cdot C) \quad \Rightarrow \quad {\omega_0}^2 = 1/(LC)$$

which gives the resonant frequency:

$$f_0 = 1 \,/\, (2\pi \cdot \sqrt{(LC)})$$

This is exactly the same expression as the cutoff frequency *fc* derived in Section 2.5. That is not a coincidence: in a second-order LC filter, the corner of the Bode plot occurs at the natural (resonant) frequency of the LC pair.

4.2 Resonant Frequency of the Crossover

Substituting the design values L = 0.566 mH and C = 4.7 μF:

$$f_0 = 1 / (2\pi \cdot \sqrt{(0.566 \times 10^{-3} \cdot 4.7 \times 10^{-6})}) \approx 3{,}086\, Hz \approx 3.09\, kHz$$

At $\omega = \omega_0$ the inductive and capacitive reactances are exactly equal in magnitude but opposite in sign, and they cancel from the impedance expression — leaving only the resistive part of the load (the loudspeaker voice coil resistance). This is the same physical mechanism that occurs in a series RLC circuit at resonance, where $Z_{in}(\omega_0) = R$.

4.3 Quality Factor Q and Selectivity

The quality factor Q measures the sharpness of the resonance peak. Higher Q means a narrower bandwidth and a stronger resonance [6]. For the second-order LPF arm of the crossover, comparison with the canonical form gives:

$$Q = R \cdot \sqrt{(C/L)}$$

Substituting the design values:

$$Q = 8 \cdot \sqrt{(4.7 \times 10^{-6} / 0.566 \times 10^{-3})} = 8 \times 0.0912 \approx 0.73$$

This Q is intentionally low (close to the Butterworth value $1/\sqrt{2} \approx 0.707$), because crossover designers want a maximally-flat response in the passband, not a sharp peak at *fc*. A high-Q crossover would produce an audible hump at the crossover point and an unpleasant tonal coloration.

The relationship between Q and bandwidth (BW) learned in Chapter 3 for second-order filters is:

$$BW = f_0 / Q$$

For the present design: $BW \approx 3{,}086 / 0.73 \approx 4.2\, kHz$, a deliberately wide skirt that ensures gentle blending between the woofer and tweeter outputs across at least an octave on either side of f_0.

4.4 Contrast with a Tuned Resonant Circuit

A resonant filter — for example, the LC tuning circuit in a radio receiver's front end — is designed for the opposite goal: maximum selectivity. Such a circuit might use L = 1 mH and C = 0.1 μF with R = 10 Ω to give f_0 = 100 kHz/(2π) ≈ 16 kHz $and\ Q = \omega_0 L/R$ = 10, exactly the worked example given in Chapter 3 [6]. With Q = 10, the bandwidth is 1.6 kHz — narrow enough to separate one radio station from its neighbour.

In the loudspeaker crossover, by contrast, we deliberately damp the LC pair down to Q ≈ 0.7 by leveraging the loudspeaker's own 8 Ω voice-coil resistance as the damping element. So the same $f_0 = 1/(2\pi\sqrt{LC})$ formula governs both circuits, but the design intent is opposite:

Property	Loudspeaker crossover	Radio tuning circuit
Goal	Smooth split between two drivers	Pick out one narrow frequency band
Q	≈ 0.7 (low — damped)	≥ 10 (high — sharp)
Bandwidth	Wide, ≈ f_0 (gentle blending)	Narrow, f_0/10 or less
Behaviour at f_0	−3 dB (handover point)	Maximum response (peak)
R chosen to be	Large (8 Ω voice-coil) → low Q	Small → high Q

Table 2 — Same f_0 formula, opposite design philosophies.

4.5 Summary of the Resonance Connection

Three points should be retained from this section:

- The cutoff frequency of a second-order LC filter is identical in form to the resonant frequency of an RLC circuit: $fc = f_0 = 1/(2\pi\sqrt{LC})$. The crossover network is, in effect, a deliberately under-damped resonant circuit operated near, but not at, its peak.
- Quality factor Q determines whether the resonance shows up as a sharp peak (high Q, narrow bandwidth, used in radio tuners) or as a smooth corner (low Q, wide bandwidth, used in audio crossovers).
- By choosing L, C, and the load R appropriately, the same circuit topology can be tuned anywhere between these two extremes, illustrating a powerful theme of AC circuit analysis: a single set of equations covers a vast range of practical applications.

References

[1] Dr. Mohammed Naim, Chapter 1: Sinusoidal Steady State Analysis — Advanced Electrical and Electronic Circuits.

[2] Dr. Mohammed Naim, Chapter 2: AC Filter Circuits — Advanced Electrical and Electronic Circuits.

[3] C. K. Alexander and M. N. O. Sadiku, Fundamentals of Electric Circuits, 6th ed., New York: McGraw-Hill, 2017, Ch. 14 (Frequency response, transfer functions, passive filters, Bode plots).

[4] V. Dickason, The Loudspeaker Design Cookbook, 7th ed., Peterborough, NH: Audio Amateur Press, 2006 — the standard industry reference for passive crossover design.

[5] S. H. Linkwitz, "Active Crossover Networks for Noncoincident Drivers," Journal of the Audio Engineering Society, vol. 24, no. 1, pp. 2–8, January/February 1976 — foundational paper on the Linkwitz–Riley alignment.

[6] Dr. Mohammed Naim, Chapter 3: RLC Circuits and Resonance — Advanced Electrical and Electronic Circuits.

[7] Crossover_networks_a_review. cross-over pdf